

QCE Specialist Mathematics Units 3&4

Paper 1 –Technology-free

Student's Name: _____

Teacher's Name: _____

Time allowed

- Perusal time – 5 minutes
- Working time – 90 minutes

General instructions

- Answer all questions in this question and response booklet.
- Calculators are **not** permitted.
- Formula booklet provided.
- Planning paper will not be marked.

Section 1 (10 marks)

- 10 multiple choice questions

Section 2 (50 marks)

- 10 short response questions

Students are advised that this is a trial examination only and cannot in any way guarantee the content or the format of the 2026 QCE Specialist Mathematics Units 3&4 examination.

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Section 1

Instructions

- This section has 10 questions and is worth 10 marks.
- Use a 2B pencil to fill in the A, B, C or D answer bubble completely.
- Choose the best answer for Questions 1–10.
- If you change your mind or make a mistake, use an eraser to remove your response and fill in the new answer bubble completely.

	A	B	C	D
Example:	☒	☐	☐	☐

	A	B	C	D
1.	☐	☐	☐	☐
2.	☐	☐	☐	☐
3.	☐	☐	☐	☐
4.	☐	☐	☐	☐
5.	☐	☐	☐	☐
6.	☐	☐	☐	☐
7.	☐	☐	☐	☐
8.	☐	☐	☐	☐
9.	☐	☐	☐	☐
10.	☐	☐	☐	☐

Ensure you have filled an answer bubble for each question.

Section 2

Instructions

- Write using black or blue pen.
- Questions worth more than one mark require mathematical reasoning and/or working to be shown to support answers.
- If you need more space for a response, use the additional pages at the back of this booklet.
 - On the additional pages, write the question number you are responding to.
 - Cancel any incorrect response by ruling a single diagonal line through your work.
 - Write the page number of your alternative/additional response, i.e. See page ...
 - If you do not do this, your original response will be marked.
- This section has 10 questions and is worth 50 marks.

QUESTION 11 (5 marks)

Consider the equation $z^4 = -8 - 8\sqrt{3}i$, $z \in \mathbb{C}$.

- a) The complex number $-8 - 8\sqrt{3}i$ can be written in the form $rcis(\theta)$.

Determine r and θ , where $-\pi < \theta \leq \pi$.

[2 marks]

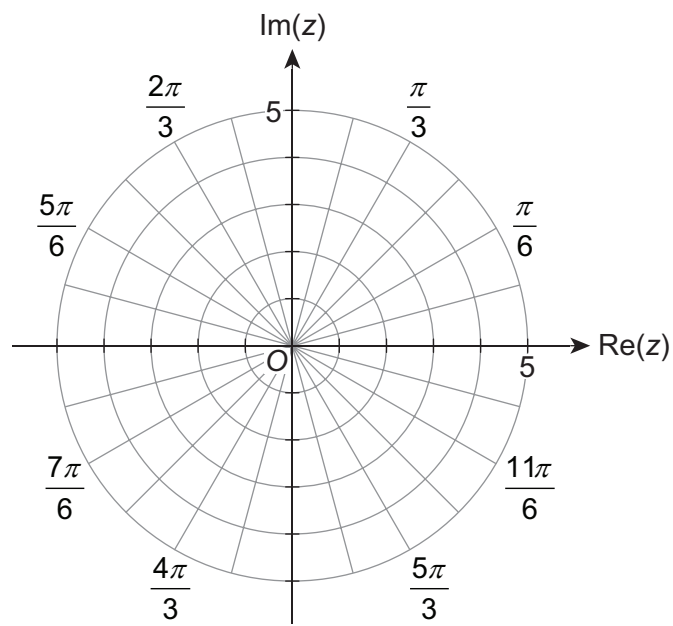
- b) One solution for $z^4 = -8 - 8\sqrt{3}i$ has a principal argument between 0 and $\frac{\pi}{2}$.

Express this solution in the form $rcis(\theta)$.

[1 mark]

- c) Plot all the solutions to $z^4 = -8 - 8\sqrt{3}i$ on the Argand diagram provided.

[2 marks]



QUESTION 12 (5 marks)

Let $f(n) = \frac{n}{3}(2n-1)(n+2)$.

It is proposed that $\sum_{i=1}^n (2i^2 - 1) = f(n)$, $\forall n \in \mathbb{Z}^+$. A student attempts to prove this statement by mathematical induction.

- a) Complete the initial step of the proof by evaluating $f(1)$ and comparing

it to $\sum_{i=1}^1 (2i^2 - 1)$.

[1 mark]

- b) In the next step of the proof, it is assumed that the statement is true for $n = k$.

Fill in the blanks in the expression to complete the assumption step of the proof. [1 mark]

$$\sum_{i=\square}^{\square} (2i^2 - 1) = f(\square)$$

- c) Use a formal process to prove that $f(k) + 2(k+1)^2 - 1 = f(k+1)$.

[3 marks]

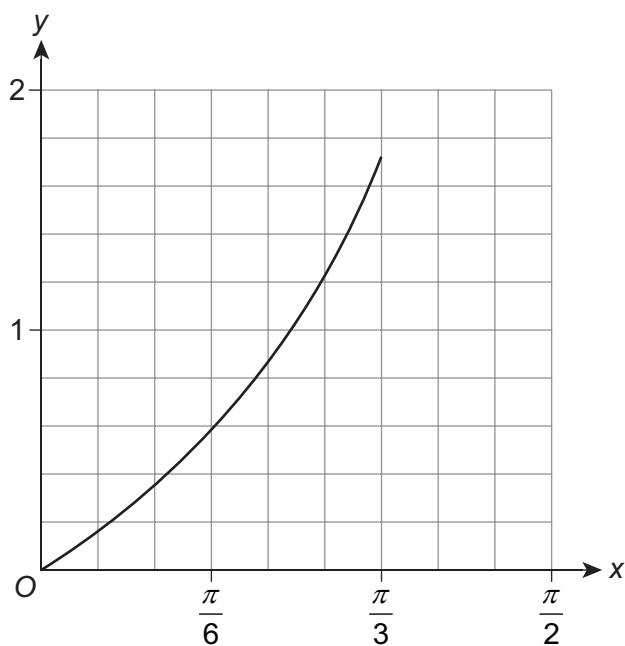
QUESTION 13 (4 marks)

The manager of a department store investigates how long customers spend inside the store. They record the time, in minutes, spent in the store for n customers, then uses this data to calculate the sample mean, $\bar{x}$, and the sample standard deviation, s . Using the sample mean, sample standard deviation and $z = 2.5$, the manager determines that the confidence interval for the mean amount of time spent in the store is (7.375, 8.625).

Given that $\bar{x} = 4s$, calculate the value of n .

QUESTION 14 (5 marks)

A region of the curve $y = \tan(x)$ is shown.



Determine the exact volume of the solid of revolution that is formed by rotating this region about the x-axis.

QUESTION 15 (5 marks)

Consider the complex number $z = a\text{cis}(A)$, where $a \in \mathbb{R}^+$ and $A = \text{Arg}(z)$.

- a) State the Cartesian form of z in terms of a and A . [1 mark]

- b) State the Cartesian form of $\bar{z}$ in terms of a and A . [1 mark]

- c) Use your results from Questions 15a) and 15b) to prove that $z\bar{z} = |z|^2$. [2 marks]

- d) State $\text{Arg}(z\bar{z})$. [1 mark]

QUESTION 16 (5 marks)

Evaluate $\int_0^1 3 \arcsin\left(\frac{x}{2}\right) dx$.

QUESTION 17 (6 marks)

Consider the equations for the line L and the plane π .

$$L: \mathbf{r} = 2\hat{\mathbf{i}} + 3\hat{\mathbf{k}} + \lambda(-2\hat{\mathbf{i}} + 3\hat{\mathbf{j}} + \hat{\mathbf{k}})$$

$$\pi: 5x + 2y + 4z = 24$$

- a) Given that $\mathbf{n}$ is a vector that is normal to plane π , determine $\hat{\mathbf{n}}$. Give your answer in the form $\frac{\sqrt{a}}{b}(x\hat{\mathbf{i}} + y\hat{\mathbf{j}} + z\hat{\mathbf{k}})$, where $a, b, x, y, z \in \mathbb{Z}$. [3 marks]

- b) Given that $\mathbf{d}$ is the direction vector of line L , calculate $\hat{\mathbf{n}} \cdot \mathbf{d}$. [1 mark]

- c) Evaluate the reasonableness of the claim that line L lies on plane π . [2 marks]

Consider the relation $x = \ln(e^{-x} + e^{-y})$.

Show that $\frac{dy}{dx} < -1$ for all points on the relation.

[illegible]

14

